Continuity

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Continuity

GENERAL INTRODUCTION :

Function is said to be continuous at x = a if while travelling along the graph of the function and in crossing over the point at x = a either from Left to Right or from Right to Left one does not have to lift pen. In case one has to lift his pen the graph of the function is said to have break or discontinuous at x=a. Different type of situations which may up at x=a along the graph can be:

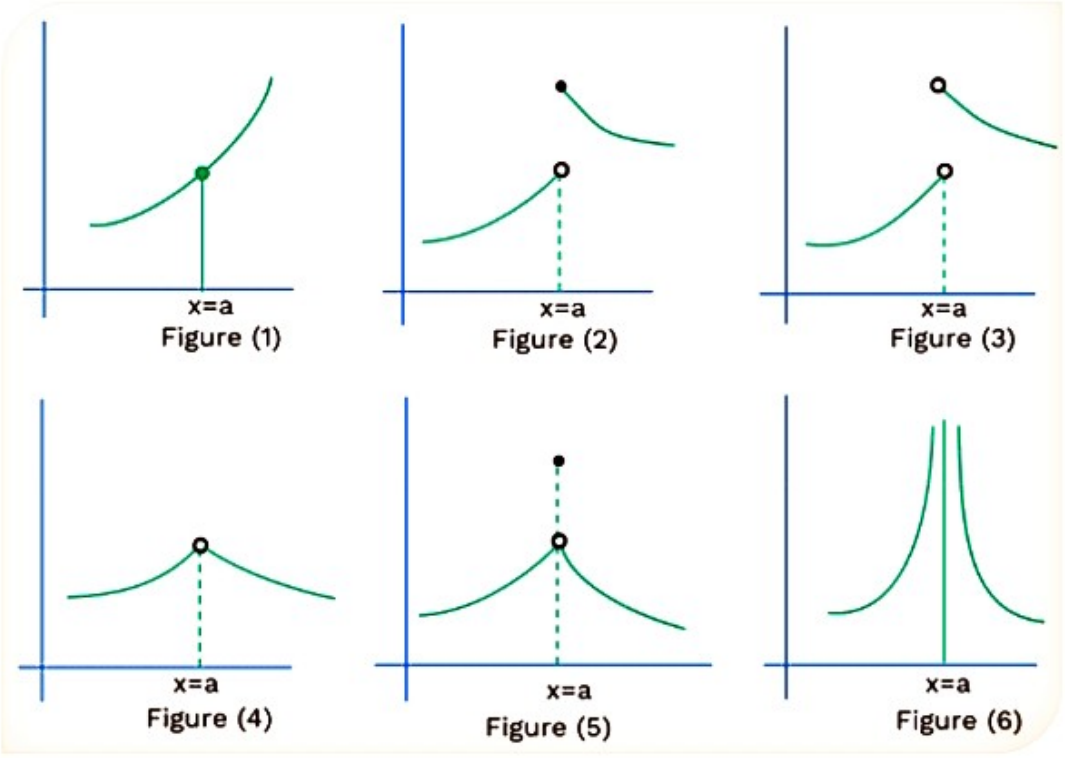


Figure (1) Figure (2) Figure (3)

Figure (4) Figure (5) Figure (6)

FORMULATIVE DEFINITION OF CONTINUITY :

A function f(x) is said to be continuous at

x = a,

If $\lim_{x\to a} f(x) = f(a)$ exists and equal to f(a).

$\Rightarrow \lim_{h\to 0} f(a-h) = \lim_{h\to 0} f(a+h) = f(a) =$ a finite quantity.

i.e. (LHL at x = a) = (RHL at x = a) = value of f(x) at x = a = a finite quantity.

NOTE :

(i) Continuity at x = a
⇒ existence of limit at x = a, but not the converse.

(ii) Continuity at x = a
⇒ f is well defined at x = a but not the converse.

(iii) Discontinuity at x = a is meaningful to talk if in the immediate neighbourhood of x = a, i.e., the function has a graph in the immediate neighbourhood of x = a, not necessarity at x = a.

(iv) Continuity is always talk in the domain of function and hence $f(x)=\frac{1}{x-1},\frac{1}{x}$, tanx are all continuous functions but if you want to talk of discontinuity then we can say $\frac{1}{x-1}$ is discontinuous at $x=1,\frac{1}{x}$ is discontinuous at x = 0.

(v) Point function are continuous.
e.g. $\sqrt{1-x}+\sqrt{x-1},\sqrt{x}+\sqrt{-x}$

(vi) Inverse of a discontinuous function can be continuous.

e.g. $f(x)=\begin{cases}1+x^2 & \text{if } x>0\\ 0 & \text{if } x=0\\ -(1+x^2) & \text{if } x<0\end{cases}$

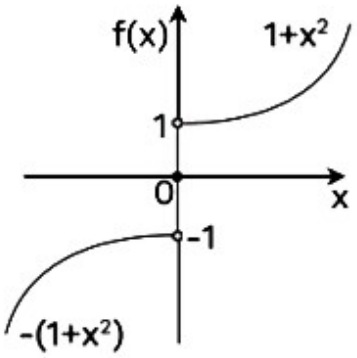
is discontinuous at x = 0 but its inverse function

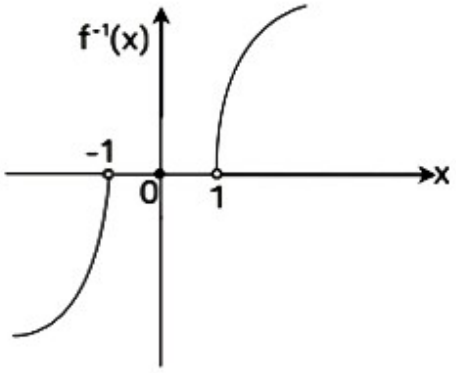
$$f^{-1}(x)=\begin{cases}\sqrt{x-1} & \text{if } x>1\\ 0 & \text{if } x=0\\ -\sqrt{-(1+x)} & \text{if } x<-1\end{cases}$$

which is a continuous function and its graph is as shown.

Point to Remember!!!

All rational functions are continuous. Because continuity is always talk in the domain of f(x).





CONTINUTY IN AN INTERVAL :

(a) A function f is said to be continuous in (a, b) if f is continuous at each and every point $\in$ (a, b).

(b) A function f is said to be continuous in a closed interval [a, b] if :

(i) f is continuous in the open interval (a,b) and

(ii) f is right continuous at **'a'**

i.e. $\lim\limits_{x\to a^+} f(x) = f(a) =$ a finite quantity.

(iii) f is left continuous at **'b'**

i.e. $\lim\limits_{x\to b^-} f(x) = f(b) =$ a finite quantity.

REASONS OF DISCONTINUITY :

A function can be discontinuous due to the following reasons.

(i)

$\lim\limits_{x\to a} f(x)$ does not exist (f(a) may or may not be defined)

i.e. $\lim\limits_{h\to 0} f(a+h) \neq \lim\limits_{h\to 0} f(a-h)$

Ex. **(a)** f(x) = [x] is discontinuous at all integer points; f(x) = sgn(x) is discontinuous at x = 0.

(b) $f(x) = \dfrac{x}{x-1}$ is discontinuous at x = 1.

(ii)

$\lim\limits_{x\to a} f(x)$ exist but is not equal to f(a) i.e.,

$\lim\limits_{h\to 0} f(a+h) = \lim\limits_{h\to 0} f(a-h) \neq f(a)$

Ex. $f(x) = \begin{cases} (1-x)\tan\dfrac{\pi x}{2} & \text{if } x \neq 1 \\ \dfrac{\pi}{2} & \text{if } x = 1 \end{cases}$

Point to Remember!!!

It should be remembered that all polynomial functions, logarithmic functions, exponential functions, trigonometric functions, inverse circular functions, and modulus functions are continuous in their domain.

$$\lim_{x\to 1} f(x) = \lim_{x\to 1}(1-x)\tan\left(\frac{\pi x}{2}\right) = \lim_{h\to 0}\frac{h}{\tan\left(\frac{\pi h}{2}\right)} = \frac{2}{\pi}$$

$\Rightarrow \lim_{x\to 1} f(x) \neq f(1) \Rightarrow f(x)$ is discontinuous at x = 1.

(iii)

f(a) is not defined

Ex. $f(x) = \frac{1}{x-1}$

Q.1 **What value must be assigned to k so that the function f(x) is continuous at x = 4?**

$$f(x) = \begin{cases} \frac{x^4 - 256}{x-4}, & x \neq 4 \\ k, & x = 4 \end{cases}$$

Sol. $$f(x) = \begin{cases} \frac{x^4 - 256}{x-4}, & x \neq 4 \\ k, & x = 4 \end{cases}$$

$$f(x) = \begin{cases} \frac{(x^2+16)(x-4)(x+4)}{(x-4)}, & x \neq 4 \\ k, & x = 4 \end{cases}$$

f(x) will be continuous for k = 256

because $\lim_{x\to 4} f(x) = 256$

Q.2 **Let** $f(x) = \begin{cases} \frac{\sin ax^2}{x^2}, & x \neq 0 \\ \frac{3}{4} + \frac{1}{4a}, & x = 0 \end{cases}$ **. For what values of a, f(x) is continuous at x = 0.**

Sol. $$f(x) = \begin{cases} \frac{\sin ax^2}{x^2}, & x \neq 0 \\ \frac{3}{4} + \frac{1}{4a}, & x = 0 \end{cases}$$

$$\lim_{x\to 0} f(x) = a = f(0)$$

$$\frac{3}{4} + \frac{1}{4a} = a$$

$$3a + 1 = 4a^2 \Rightarrow 4a^2 - 3a - 1 = 0$$

$$\Rightarrow 4a^2 - 4a + a - 1 = 0$$

$$\Rightarrow 4a(a-1) + 1(a-1) = 0$$

$$\Rightarrow a = 1, -\frac{1}{4}$$

Q.3 $f(x) = \begin{cases} (\cos x)^{\cot^2 x} & x \neq 0 \\ e^{\frac{-1}{2}} & \text{if} \quad x = 0 \end{cases}$ **find whether f(x) is continuous at x = 0 or not.**

Sol. $\lim_{x\to 0} f(x) = \lim_{x\to 0} (\cos x)^{\cot^2 x}$

$= e^{\lim_{x\to 0}(\cos x - 1)\cot^2 x} = e^{\lim_{x\to 0} \frac{-(1-\cos x)}{x^2} \frac{x^2}{\tan^2 x}} = e^{\frac{-1}{2}} = f(0) \Rightarrow f(x)$ is continuous at x = 0.

Q.4 **If** $f(x) = \begin{cases} \dfrac{(e^x - 1)^3 \operatorname{cosec}(x)}{\ln(1+x^2)} & x \neq 0 \\ b & x = 0 \end{cases}$ **is continuous, find b.**

Sol. $\lim_{x\to 0} \dfrac{(e^x - 1)^3 \operatorname{cosec}(x)}{\ln(1+x^2)} = \lim_{x\to 0} \dfrac{\left(\dfrac{e^x - 1}{x}\right)^3}{\dfrac{\ln(1+x^2)}{x^2}} \cdot \left(\dfrac{x}{\sin x}\right) = 1 = f(0)$

$\therefore$ b = 1.

Q.5 **Find the values of 'a' and 'b' so that the function**

$$f(x) = \begin{cases} x + a\sqrt{2}\sin x & 0 \le x < \frac{\pi}{4} \\ 2x\cot x + b & \frac{\pi}{4} \le x < \frac{\pi}{2} \\ a\cos 2x - b\sin x & \frac{\pi}{2} \le x < \pi \end{cases}$$ **is continuous in [0, π]**

Sol. f(x) is continuous in the interval $0 \le x < \frac{\pi}{4}, \frac{\pi}{4} < x < \frac{\pi}{2}, \frac{\pi}{2} < x \le \pi$.

We need to make the function continuous at $x = \frac{\pi}{4}, \frac{\pi}{2}$

For continuity at $x = \frac{\pi}{4}$, $\lim\limits_{x \to \left(\frac{\pi}{4}\right)^-} f(x) = \lim\limits_{x \to \left(\frac{\pi}{4}\right)^+} f(x) = f\left(\frac{\pi}{4}\right)$

$$\lim_{x \to \left(\frac{\pi}{4}\right)^-} \left(x + a\sqrt{2}\sin x\right) = \lim_{x \to \left(\frac{\pi}{4}\right)^+} (2x \cot x + b) = f\left(\frac{\pi}{4}\right)$$

$$\Rightarrow \quad \frac{\pi}{4} + a\sqrt{2}\sin\left(\frac{\pi}{4}\right) = 2.\frac{\pi}{4}.\cot\left(\frac{\pi}{4}\right) + b = 2\frac{\pi}{4}.\cot\left(\frac{\pi}{4}\right) + b$$

$$\Rightarrow \quad \frac{\pi}{4} + a = \frac{\pi}{2} + b \Rightarrow a - b = \frac{\pi}{4} \qquad \ldots(1)$$

For continuity at $x = \frac{\pi}{2}$, $\lim\limits_{x \to \left(\frac{\pi}{2}\right)^-} f(x) = \lim\limits_{x \to \left(\frac{\pi}{2}\right)^+} f(x) = f\left(\frac{\pi}{2}\right)$

$$\Rightarrow \quad \lim_{x \to \left(\frac{\pi}{2}\right)^+} (2x \cot x + b) = \lim_{x \to \left(\frac{\pi}{2}\right)^+} (a\cos 2x - b\sin x) = a\cos\pi - b\sin\frac{\pi}{2}$$

$$\Rightarrow \quad 0 + b = -a - b \Rightarrow a + 2b = 0 \qquad \ldots(2)$$

From equation (1) and (2)

$$a = \frac{\pi}{6}, b = \frac{-\pi}{12}.$$

Q.6 $f(x) = \begin{cases} \dfrac{1-\cos 4x}{x^2} & \text{if} \quad x < 0 \\ a & \text{if} \quad x = 0 \\ \dfrac{\sqrt{x}}{\sqrt{16+\sqrt{x}} - 4} & \text{if} \quad x > 0 \end{cases}$ **Determine 'a' if possible so that the function is continuous at x = 0.**

Sol. $f(0^-) = \lim\limits_{x \to 0^-} f(x) = \lim\limits_{x \to 0^-} \dfrac{1 - \cos 4x}{x^2} = 8$

$$f(0^+) = \lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} \frac{\sqrt{x}}{\sqrt{16+\sqrt{x}} - 4} = \lim_{x \to 0^+} \frac{\sqrt{x}.\left(\sqrt{16+\sqrt{x}} + 4\right)}{16 + \sqrt{x} - 16}$$

$$= \lim_{x \to 0^+} \left(\sqrt{16} + \sqrt{x} + 4\right) = 8$$

$$f(0^-) = f(0^+) = 8 = f(0) \Rightarrow a = 8$$

Q.7 $f(x)=\begin{cases}\left(1+|\sin x|\right)^{\frac{a}{|\sin x|}} & \text{for} \quad -\frac{\pi}{6}<x<0\\ b & \text{for} \quad x=0\\ e^{\frac{\tan 2x}{\tan 3x}} & \text{for} \quad 0<x<\frac{\pi}{6}\end{cases}$ **Find 'a' and 'b' if f is continuous at x = 0.**

Sol. $\lim\limits_{x\to 0^-} f(x)=\lim\limits_{x\to 0}\left(1+|\sin x|\right)^{\frac{a}{|\sin x|}}=e^{a}$

$\lim\limits_{x\to 0^+} f(x)=\lim\limits_{x\to 0^+} e^{\frac{\tan 2x}{\tan 3x}}=e^{\lim\limits_{x\to 0^+}\frac{\tan 2x}{2x}\cdot\frac{3x}{\tan 3x}\cdot\frac{2}{3}}=e^{\frac{2}{3}}$

$\therefore \quad e^{a}=b=e^{\frac{2}{3}}\Rightarrow a=\frac{2}{3}, b=e^{\left(\frac{2}{3}\right)}$

Q.8 **Let** $f(x)=\begin{cases}\dfrac{\left(e^{2x}+1\right)-(x+1)\left(e^{x}+e^{-x}\right)}{x\left(e^{x}-1\right)} & \text{if} \quad x\neq 0\\ k & \text{if} \quad x=0\end{cases}$

if f(x) is continuous at x = 0 then k is equal to

(A) $\frac{1}{2}$ **(B) 1** **(C)** $\frac{3}{2}$ **(D) 2**

Sol. **(D)**

$$k=\lim_{x\to 0}\frac{\left(e^{2x}+1\right)-(x+1)\left(e^{x}+e^{-x}\right)}{x\left(e^{x}-1\right)}$$

By L Hospital Rule

$$=\lim_{x\to 0}\frac{2e^{2x}-(x+1)\left(e^{x}-e^{-x}\right)-\left(e^{x}+e^{-x}\right)}{\left(e^{x}-1\right)+xe^{x}}$$

By L Hospital Rule again

$$=\lim_{x\to 0}\frac{4.e^{2x}-(x+1)\left(e^{x}+e^{-x}\right)-\left(e^{x}-e^{-x}\right)-\left(e^{x}-e^{-x}\right)}{e^{x}+\left(x\cdot e^{x}+1\cdot e^{x}\right)}=\frac{4-(2)-0-0}{1+1}=1$$

Q.9 **If** $f(x)=\begin{bmatrix} x+1 & x\le 1 \\ 3-ax^2 & x>1 \end{bmatrix}$ **is continuous at x = 1, then find the value of a.**

Sol. $f(x)=\begin{bmatrix} x+1 & x\le 1 \\ 3-ax^2 & x>1 \end{bmatrix}$

$\lim\limits_{x\to 1^-} f(x)=\lim\limits_{x\to 1^+} f(x)=f(1)$

$2=3-a \Rightarrow a=1$

Q.10 **Let** $f(x)=\dfrac{\sqrt{x^2+kx+1}}{x^2-k}$**. The interval(s) of all possible values of k for which f is continuous for every** $x\in R$**, is**

(A) $(-\infty,-2)$ **(B)** $[-2,0)$ **(C)** $R-(-2,2)$ **(D)** $(-2,2)$

Sol. **(B)**

$x^2-k\neq 0 \ \forall \ x\in R$

$\Rightarrow \quad k<0$ ...(1)

$x^2+kx+1\ge 0 \ \forall \ x\in R$

$\Rightarrow \quad k^2-4\le 0 \Rightarrow -2\le k\le 2$ ...(2)

$k\in[-2,0)$

Q.11 **The function f : R-{0} →R given by** $f(x)=\dfrac{1}{x}-\dfrac{2}{e^{2x}-1}$**, is continuous at x = 0, then find the value of f(0).**

Sol. $\lim\limits_{x\to 0} f(x)=f(0)$

$=\lim\limits_{x\to 0}\dfrac{1}{x}-\dfrac{2}{e^{2x}-1}=\lim\limits_{x\to 0}\dfrac{1}{x}-\dfrac{1}{x(1+x+\ldots)}$

$=\lim\limits_{x\to 0}\dfrac{1}{x}\left[\dfrac{1+x+\ldots-1}{1+x+\ldots}\right]=1=f(0)$

Q.12 **Let** $f(x)=\begin{cases}(1+3x)^{\frac{1}{x}}, & x\neq 0 \\ e^3, & x=0\end{cases}$**. Discuss the continuity of f(x) at**

(i) x = 0, **(ii) x = 1.**

Sol. **(i)** $\lim\limits_{x\to 0} f(x) = \lim\limits_{x\to 0}(1+3x)^{\frac{1}{x}} = e^{\lim\limits_{x\to 0}\frac{1}{x}(1+3x-1)} = e^3 = f(0)$

hence continuous at x = 0

(ii) $\lim\limits_{x\to 1} f(x) = 4 = f(1)$

hence continuous at x = 1

Q.13 $f(x) = \begin{cases} \dfrac{\sin x + \sin 5x}{\cos x + \cos 5x} & \text{if } x \neq -\dfrac{\pi}{4} \\ k & \text{if } x = -\dfrac{\pi}{4} \end{cases}$ **. Find k if f is continuous at** $x = -\dfrac{\pi}{4}$

(A) 1 **(B) –1** **(C) 0** **(D)** $-\dfrac{1}{\sqrt{2}}$

Sol. **(A)**

$$f(x) = \begin{cases} \dfrac{\sin x + \sin 5x}{\cos x + \cos 5x} & \text{if } x \neq -\dfrac{\pi}{4} \\ k & \text{if } x = -\dfrac{\pi}{4} \end{cases}$$

$$k = \lim_{x\to\frac{-\pi}{4}} \frac{\sin x + \sin 5x}{\cos x + \cos 5x}$$

$$k = \lim_{x\to\frac{-\pi}{4}} \frac{2\sin 3x\cos 2x}{2\cos 3x\cos 2x} = 1$$

Q.14 **Let** $f(x) = \begin{cases} \dfrac{(x^2+3x-1)\tan x}{x^2+2x} & \text{if } x \neq 0 \\ k & \text{if } x = 0 \end{cases}$ **; then find the value of k if f(x) is continuous at x = 0.**

Sol. $f(x) = \begin{cases} \dfrac{(x^2+3x-1)\tan x}{x^2+2x} & \text{if } x \neq 0 \\ k & \text{if } x = 0 \end{cases}$;

$$k = \lim_{x\to 0} \frac{(x^2+3x-1)}{(x+2)x}\tan x$$

so, $k = \dfrac{-1}{2}$

Q.15 **Which of the following functions is not continuous $\forall\ x \in R$?**

(A) $\sqrt{2\sin x + 3}$ **(B)** $\dfrac{e^x + 1}{e^x + 3}$ **(C)** $\left(\dfrac{2^{2x} + 1}{2^{3x} + 5}\right)^{\frac{5}{7}}$ **(D)** $\sqrt{\text{sgn}x + 1}$

Sol. **(D)**

(A) $2\sin x + 3$ is always defined. Hence continuous

(B) $\dfrac{e^x + 1}{e^x + 3} = \dfrac{e^x + 3}{e^x + 3} - \dfrac{2}{e^x + 3} = 1 - \dfrac{2}{e^x + 3}$

Which is always defined, hence continuous,

(C) $\left(\dfrac{2^{2x} + 1}{2^{3x} + 5}\right)^{\frac{5}{7}}$

Always defined and continuous.

(D) $\sqrt{\text{sgn}x + 1}$

$\text{sgn}x + 1 \geq 0$

$\text{sgn}x \geq -1 \Rightarrow \text{sgn}x = 0, 1, -1$ that is discontinuous at $x = 0$.

Q.16 **Let** $f(x) = \begin{cases} \dfrac{a + 3\cos x}{x^2}, & x \neq 0 \\ b\tan\left(\dfrac{\pi}{[x + 3]}\right), & x = 0 \end{cases}$ **. If f(x) is continuous at x = 0, then find a and b, where [.] denotes the greatest integer function.**

Sol. $f(x) = \begin{cases} \dfrac{a + 3\cos x}{x^2}, & x \neq 0 \\ b\tan\left(\dfrac{\pi}{[x + 3]}\right), & x = 0 \end{cases}$

$\lim\limits_{x \to 0} f(x) = f(0) = b\sqrt{3}$

$\lim\limits_{x \to 0} \dfrac{a + 3\cos x}{x^2} = b\sqrt{3}$ $\Rightarrow$ for $a = -3$ we get

$\lim\limits_{x \to 0} \dfrac{(-3)\left(2\sin^2 \dfrac{x}{2}\right)}{4\dfrac{x^2}{4}} = b\sqrt{3} \Rightarrow -\dfrac{3}{2} = b\sqrt{3}$

$b = -\dfrac{\sqrt{3}}{2}$ and $a = -3$

Q.17 **Discuss the continuity of** $f(x)=\begin{cases}\dfrac{x^4-5x^2+4}{|(x-1)(x+2)|}, & x\neq 1,-2\\ 6, & x=1\\ 12, & x=-2\end{cases}$

Sol. $\lim\limits_{x\to 1^-} f(x)=\lim\limits_{x\to 1^-}\dfrac{(x^2-1)(x^2-4)}{|(x-1)(x+2)|}=\lim\limits_{x\to 1^-}\dfrac{(x-1)(x+1)(x-2)(x+2)}{(x-1)(x+2)}$

$=-2\neq f(1)$

hence discontinuous.

$\lim\limits_{x\to -2^+} f(x)=\lim\limits_{x\to -2^+}\dfrac{(x-1)(x+1)(x-2)(x+2)}{-(x-1)(x+2)}=-4\neq f(-2)$

hence discontinuous.

Q.18 **Find the points of discontinuity of the following functions.**

(i) $f(x)=\dfrac{1}{2\sin x-1}$

Sol. $f(x)=\dfrac{1}{2\sin x-1}$

f(x) is discontinuous when $2\sin x-1=0$

$\Rightarrow \sin x=\dfrac{1}{2}\Rightarrow x=2n\pi+\dfrac{\pi}{6}$ or $x=2n\pi+\dfrac{5\pi}{6}$, $n\in Z$

Q. **(ii)** $f(x)=\dfrac{1}{x^2-3|x|+2}$

Sol. $f(x)=\dfrac{1}{x^2-3|x|+2}$

f(x) is discontinuous when $x^2-3|x|+2=0$

$\Rightarrow |x|^2-3|x|+2=0$

$\Rightarrow (|x|-1)(|x|-2)=0\Rightarrow |x|=1,2\Rightarrow x=\pm 1,\pm 2$

Q. **(iii)** $f(x)=\dfrac{1}{x^4+x^2+1}$

Sol. $f(x)=\dfrac{1}{x^4+x^2+1}=\dfrac{1}{\left(x^2+\frac{1}{2}\right)^2+\frac{3}{4}}$

Now, $x^4+x^2+1=\left(x^2+\dfrac{1}{2}\right)^2+\dfrac{3}{4}\geq 1\forall\ x\in R$

$\Rightarrow$ f(x) is continuous $\forall\ x\in R$

Q. **(iv)** $f(x)=\dfrac{1}{1-e^{\left(\frac{x-1}{x-2}\right)}}$

Sol. $f(x)=\dfrac{1}{1-e^{\left(\frac{x-1}{x-2}\right)}}$

f(x) is discontinuous when x - 2 = 0, also

when $1-e^{\left(\frac{x-1}{x-2}\right)}=0$

$\Rightarrow$ x = 2 and $e^{\left(\frac{x-1}{x-2}\right)}=1$

$\Rightarrow$ x = 2 and $\dfrac{x-1}{x-2}=0$

$\Rightarrow$ x = 2 and x = 1

Q. **(v)** $f(x)=[[x]]-[x-1]$, **where [.] represents the greatest integer function.**

Sol. $f(x)=[[x]]-[x-1]=[x]-([x]-1)=1$

$\Rightarrow$ f(x) is continuous $\forall\ x\in R$.

Q.19 **If** $f(x+y)=f(x).f(y)$ **for all x and y** $\in$ **R and** $f(x)=1+g(x)G(x)$, **where** $\lim\limits_{x\to 0} g(x)=0$ **and** $\lim\limits_{x\to 0}G(x)$ **exist. Prove that f(x) is continuous for all x.**

Sol. Let x = a, where 'a' is any real number.

$\lim\limits_{x\to a^-} f(x)=\lim\limits_{h\to 0} f(a-h)=\lim\limits_{h\to 0} f(a)f(-h)$

$=f(a)\left(1+\lim\limits_{h\to 0}\ g(-h)\ G(-h)\right)=f(a)$

Similarly, $\lim\limits_{x\to a^+} f(x)=f(a)=\lim\limits_{x\to a^-} f(x)$ so f(x) is continuous at x = a $\forall$ a $\in$ R.

Q.20 **If** $f(x)=\begin{cases}\dfrac{8^x-4^x-2^x+1}{x^2}, & x>0\\ e^x\sin x+4x+k\ln 4, & x\le 0\end{cases}$ **is continuous at x = 0, then find the value of k.**

Sol. $f(x)=\begin{cases}\dfrac{8^x-4^x-2^x+1}{x^2}, & x>0\\ e^x\sin x+4x+k\ln 4, & x\le 0\end{cases}$

$$\lim_{x\to 0^+}\frac{8^x-4^x-2^x+1}{x^2}=\lim_{x\to 0^+}\frac{e^{x\ln_e 8}-e^{x\ln_e 4}-e^{x\ln_e 2}+1}{x^2}$$

$$=\lim_{x\to 0^+}\frac{\left(1+x\ln 8+\frac{(x\ln 8)^2}{2!}+\ldots\right)-\left(1+x\ln 4+\frac{(x\ln 4)^2}{2!}+\ldots\right)-\left(1+x\ln 2+\frac{(x\ln 2)^2}{2!}+\ldots\right)+1}{x^2}.$$

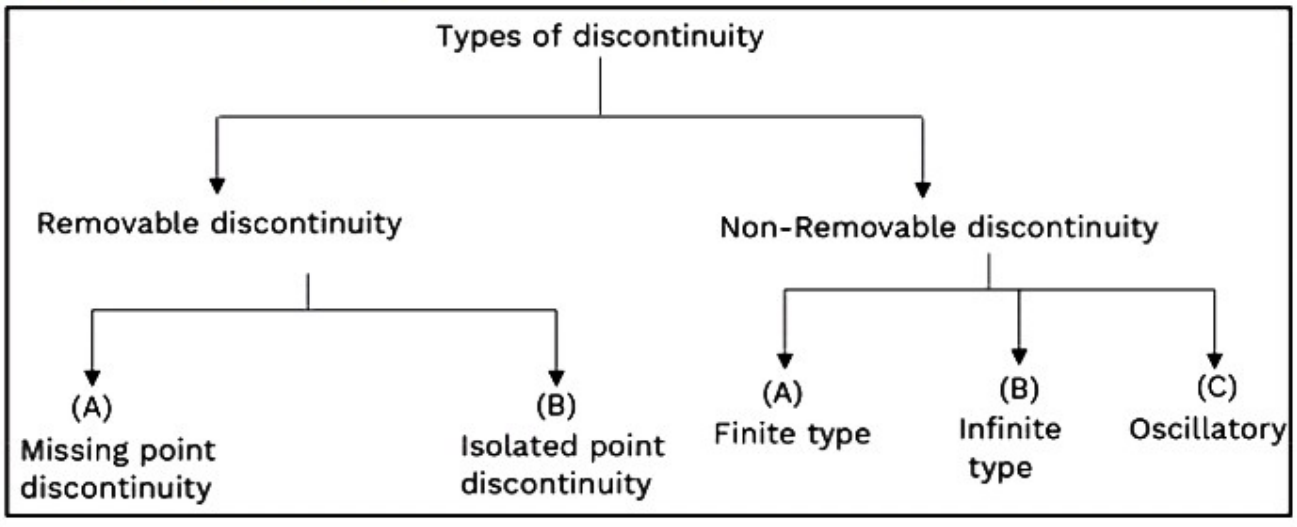
$$=\lim_{x\to 0^+}\frac{x(\ln 8-\ln 4-\ln 2)+x^2\left(\frac{(\ln 8)^2}{2}-\frac{(\ln 4)^2}{2}-\frac{(\ln 2)^2}{2}\right)}{x^2}$$

$= 2(\ln 2)^2$

$\lim_{x\to 0^-} f(x) = 0 + 0 + k\ln 4 = (\ln 2)(\ln 4)$

$\Rightarrow k = \ln 2.$

TYPES OF DISCONTINUITY :



REMOVABLE DISCONTINUITY :

Here $\lim_{x \to a} f(x)$ necessarily exists, but is either not equal to f(a) or f(a) is not defined. In this case, therefore it is possible to redefine the function in such a manner that $\lim_{x \to a} f(x) = f(a)$ and thus making the function continuous. These discontinuities can be further classified as

(a) Missing point discontinuity :

Here $\lim_{x \to a} f(x)$ exists. But f(a) is not defined.

(b) Isolated point discontinuity :

Here $\lim_{x \to a} f(x)$ exists, also f(a) is defined but $\lim_{x \to a} f(x) \neq f(a)$

NON-REMOVABLE DISCONTINUTY

Here $\lim_{x \to a} f(x)$ does not exists and therefore it is not possible to redefine the function in any manner to make it continuous. Such discontinuities can be further classified into three fold.

(a) Finite type :

Both limits finite and unequal.

In this case non-negative difference between the two limits is called the Jump of discontinuity. A function having a finite number of jumps in a given interval I is called a Piece Wise Continuous or Sectionally Continuous function in this interval.

(b) Infinite type :

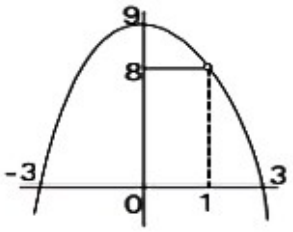
At least one of the two limit are infinity.

(c) Oscillatory :

Limits oscillate between two finite quantities.

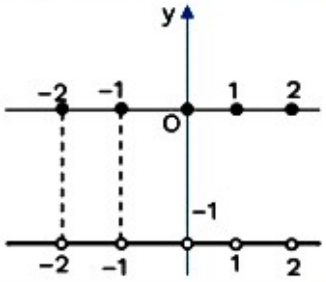
Q.1 **Discuss the nature of discontinuity for** $f(x)=\dfrac{(x-1)(9-x^2)}{x-1},\ x\neq 1$

Sol. At x = 1, f(1) is not defined.
Hence f(x) has missing point of discontinuity at x = 1.



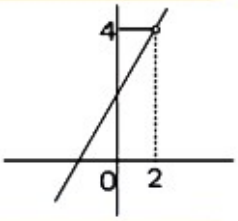
Q.2 **Discuss the nature of discontinuity for** $f(x)=[x]+[-x]=\begin{cases}0 & \text{if } x\in I\\ -1 & \text{if } x\notin I\end{cases}$

Sol. f(x) has isolated point of discontinuity at all integral points.



Q.3 **Discuss the nature of discontinuity for** $f(x)=\dfrac{x^2-4}{x-2},\ x\neq 2$

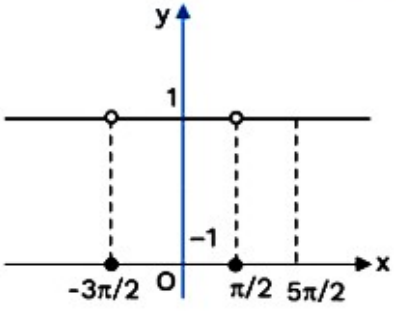
Sol. f(2) is not defined. Hence, f(x) has missing point of discontinuity at x = 2



Q.4 **Discuss the nature of discontinuity for**

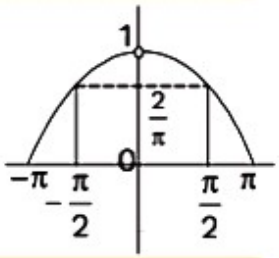
$$f(x)=\operatorname{sgn}(\cos 2x-2\sin x+3)=\operatorname{sgn}\big(2(2+\sin x)(1-\sin x)\big)=\begin{cases}0 & \text{if } x=2n\pi+\dfrac{\pi}{2}\\ +1 & \text{if } x\neq 2n\pi+\dfrac{\pi}{2}\end{cases}$$

Sol. f(x) has an isolated point discontinuity at $x=2n\pi+\dfrac{\pi}{2}$



Q.5 **Discuss the nature of discontinuity for** $f(x)=\frac{\sin x}{x}, x \neq 0$

Sol. f(0) is not defined. f(x) has missing point of discontinuity at x = 0.



Q.6 **Discuss the nature of discontinuity for** $\frac{|\sin x|}{x}$ **at x = 0.**

Sol. $\lim\limits_{x\to 0}\frac{|\sin x|}{x}$ $\begin{cases} f(0^+)=1 \\ f(0^-)=-1 \end{cases}$; jump = 2

Hence, it has finite type non removable discontinuity.

Q.7 **Discuss the nature of discontinuity for** $f(x)=\sin\frac{1}{x}$ **and** $f(x)=\cos\frac{1}{x}$ **at x = 0.**

Sol. $f(x)=\sin\frac{1}{x}$ and $f(x)=\cos\frac{1}{x}$ at x = 0 oscillates between – 1 and 1.

Hence, it has oscillatory non removable discontinuity.

Q.8 **Discuss the nature of discontinuity for** $\frac{[x]}{x}$ **at x = 2.**

Sol. $\lim\limits_{x\to 2}\frac{[x]}{x}$ $\begin{cases} f(2^+)=1 \\ f(2^-)=\frac{1}{2} \end{cases}$; jump = $\frac{1}{2}$

Hence, it has finite type non removable discontinuity.

Q.9 **Discuss the nature of discontinuity for** $f(x)=\frac{x}{1-x}$ **at x = 1.**

Sol. $f(x)=\frac{x}{1-x}$ at x=1 $\begin{cases} f(1^+)=-\infty \\ f(1^-)=+\infty \end{cases}$

Hence, it has infinite type non removable discontinuity.

Q.10 **Discuss the nature of discontinuity for $f(x)=\left[1+\frac{1}{3}\sin\left(\ln|x|\right)\right]$ at x = 0, where [.] represents greatest integer function.**

Sol. $f(x)=\left[1+\frac{1}{3}\sin\left(\ln|x|\right)\right]$ at x = 0 oscillates between 0 and 1.

Hence, it has oscillatory non removable discontinuity.

Q.11 **Discuss the nature of discontinuity for $f(x)=2^{\tan x}$ at $x=\frac{\pi}{2}$.**

Sol. $f(x)=2^{\tan x}$ at $x=\frac{\pi}{2}$ $\begin{cases} f\left(\frac{\pi^+}{2}\right)=0 \\ f\left(\frac{\pi^-}{2}\right)=\infty \end{cases}$

Hence, it has infinite type non removable discontinuity.

Q.12 **Discuss the nature of discontinuity for $f(x)=\frac{1}{x^2}$ at x = 0.**

Sol. $f(x)=\frac{1}{x^2}$ at x = 0 $\begin{cases} f(0^+)=\infty \\ f(0^-)=\infty \end{cases}$

Hence, it has infinite type non removable discontinuity.

Q.13 **Discuss the nature of discontinuity for the following functions :**

(i) $f(x)=\frac{1}{x-1}$

Sol. $f(x)=\frac{1}{x-1}$

x = 1 is the vertical asymptote and hence has non-removable discontinuity.

Q. **(ii)** $f(x)=\frac{x^3-x}{x^2-1}$

Sol. $f(x)=\frac{x^3-x}{x^2-1}=\frac{x\left(x^2-1\right)}{x^2-1}$

If f(1) = 1 and f(−1) = −1 then f(x) is continuous.

Hence, removable discontinuity and missing point discontinuity.

Q. **(iii)** $f(x)=\frac{|x-1|}{x-1}$

Sol. $f(x)=\frac{|x-1|}{x-1}$

$\lim_{x\to 1^-} f(x) \neq \lim_{x\to 1^+} f(x)$

Hence, non-removable discontinuity and also jump of discontinuity.

Q. **(iv)** $f(x)=\sin\left(\frac{1}{x-1}\right)$

Sol. $f(x)=\sin\left(\frac{1}{x-1}\right)$

At x=1,

$\lim_{x\to 1} f(x)$ oscillates between − 1 to 1.

Hence, non-removable discontinuity and oscillating discontinuity.

Q.14 **State the number of points of discontinuities and discuss the nature of discontinuity for the function** $f(x)=\frac{1}{\ln|x|}$.

Sol. $f(x)=\begin{cases}\frac{1}{\ln x} & \text{if } x>0, \quad x\neq 1\\ \frac{1}{\ln(-x)} & \text{if } x<0, \quad x\neq -1\end{cases}$ function is obviously discontinuous at x = 0, 1, −1, as it is not defined.

$\left.\begin{matrix}\lim_{x\to 0^+} f(x)=0\\ \lim_{x\to 0^-} f(x)=0\end{matrix}\right\}$ Limit exists at x=0. Hence there exists removable discontinuity at x = 0. (Missing point discontinuity)

$\left.\begin{array}{l}\lim\limits_{x\to 1^+} f(x) = \infty \\ \lim\limits_{x\to 1^-} f(x) = -\infty\end{array}\right\}$ Limit DNE. Hence non removable discontinuity (infinite type) x = 1

$\left.\begin{array}{l}\lim\limits_{x\to -1^+} f(x) = -\infty \\ \lim\limits_{x\to -1^-} f(x) = \infty\end{array}\right\}$ Limit DNE.

Hence non removable discontinuity (infinite type) at x = –1

THEOREMS ON CONTINUITY :

T-1 :

Sum, difference, product and quotient of two continuous functions is always a continuous function.

However, $h(x) = \dfrac{f(x)}{g(x)}$ is continuous at x = a

only if g(a) ≠ 0.

IMPORTANT NOTES :

(a)

If f(x) is continuous and g(x) is discontinuous then f(x) + g(x) is a discontinuous function.

Proof :

Let f(x) + g(x) is a continuous function.

so, $\lim\limits_{x\to a}\left(f(x)+g(x)\right) = f(a)+g(a)$ ...(1)

Also, f(x) is a continuous function $\lim\limits_{x\to a} f(x) = f(a)$ ...(2)

From (1) and (2)

$\lim\limits_{x\to a} g(x) = g(a) \Rightarrow$ g(x) is continuous at x = a.

But given g(x) is discontinuous at x = a.

This is a contradiction.

Hence our assumption is wrong.

$\Rightarrow$ f(x) + g(x) will be discontinuous function.

(b)

If f(x) is continuous & g(x) is discontinuous at x = a then product function $\phi(x) = f(x).\, g(x)$ is not necessarily discontinuous at x = a.

Q.1 **If** $f(x) = x$ & $g(x) = \begin{cases} \sin\frac{1}{x} & x \neq 0 \\ 0 & x = 0 \end{cases}$ **then discuss the continuity for f(x)·g(x).**

Sol. $f(x)\cdot g(x) = \begin{cases} x\sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$ is continuous at x = 0.

Q.2 **If** $f(x) = \cos\left(\frac{2x-1}{2}\right)\pi$ **and g(x) = [x] ([.] denotes the greatest integer function) then discuss the continuity of f(x)·g(x).**

Sol. $f(x) = \cos\left(\frac{2x-1}{2}\right)\pi$ is continuous at x = 1.

g(x) = [x] is discontinuous at x = 1.

$\lim\limits_{x\to 1^+} \cos\left(\frac{2x-1}{2}\right)\pi.[x] = \cos\frac{\pi}{2}.(1) = 0$

$\lim\limits_{x\to 1^-} \cos\left(\frac{2x-1}{2}\right)\pi.[x] = \cos\frac{\pi}{2}.(0) = 0$

$f(1) = 0$

$\Rightarrow$ f(x)·g(x) is continuous at x = 1.

(c)

If f(x) and g(x) both are discontinuous at x = a then the product function $\phi(x) = f(x).g(x)$ is not necessarily discontinuous at x = a.

E.g. $f(x) = -g(x) = \begin{cases} 1 & x \geq 0 \\ -1 & x < 0 \end{cases}$

$\therefore$ f(x) g(x) = –1 $\forall$ x$\in$ R which is continuous function.

Q.3 If $f(x)=\begin{cases}|x+1|; & x<0\\ x; & x\geq 0\end{cases}$ and $g(x)=\begin{cases}|x+1|; & x<0\\ -|x+2|+3; & x\geq 0\end{cases}$ **discuss continuity of f(x) + g(x).**

Sol. Since f(x) is discontinuous at x = 0 and g(x) is continuous at x = 0, then f(x) + g(x) is discontinuous at x = 0.

At all other places, both functions are continuous.
Hence their sum will be continuous ⇒ f(x) + g(x) is discontinuous at x = 0 only.

T-2 : Intermediate value theorem :

If f is continuous on [a, b] and f(a) ≠ f(b) then for any value c ∈ (f(a), f(b)), there is at least one number x_0 in (a, b) for which $f(x_0) = c$.

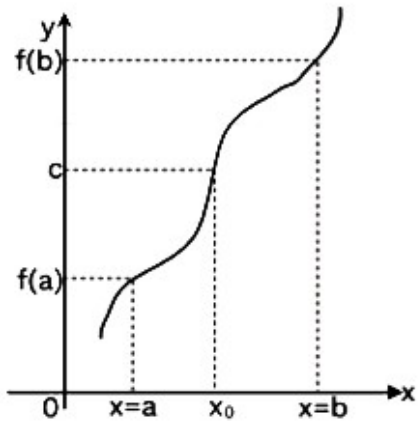


Figure (i)

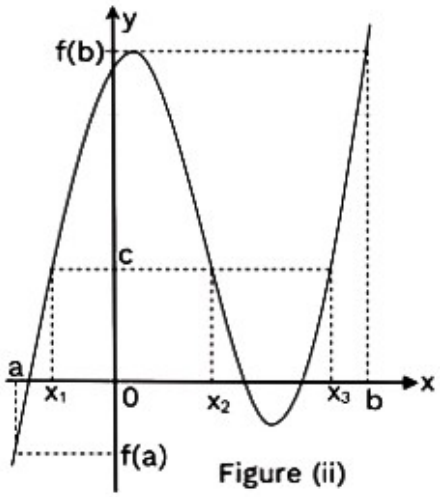


Figure (ii)

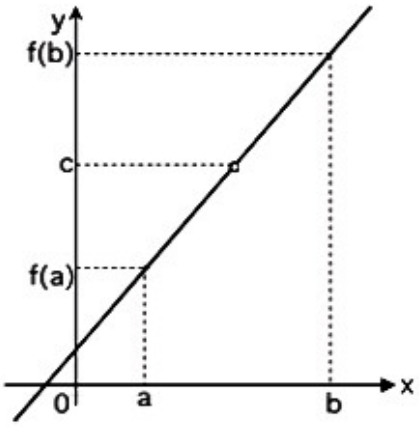


Figure (iii)

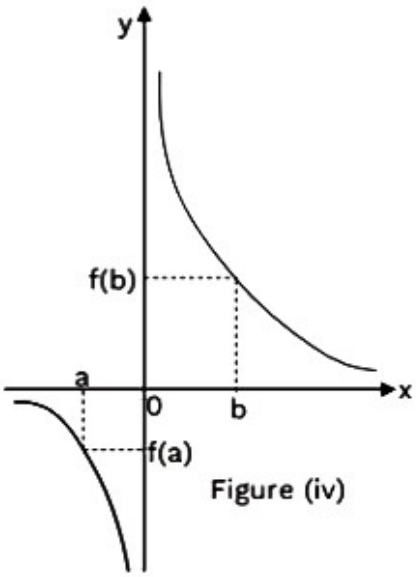


Figure (iv)

Note :

(1) Continuity through the interval [a, b] is essential for the validity of this theorem.

(2) In figure-(iv), f(a) and f(b) are of opposite sign but f(x) has no root in (a, b) as f is discontinuous.

Q.1 **Show that the function** $f(x) = (x-a)^2(x-b)^2 + x$ **takes the value** $\frac{a+b}{2}$ **for some value of** $x \in [a,b]$.

Sol. $f(x) = (x-a)^2(x-b)^2 + x$; as f(x) is continuous on [a, b] and f(a)= a and f(b) = b, then for any value $c \in (a, b)$, there is at least one number x_0 in (a, b) for which $f(x_0) = c = \frac{a+b}{2}$.

Q.2 **Suppose that f(x) is continuous in [0, 1] and f(0) = 0, f(1) = 0. Prove that f(c) = 1 - 2c² for some c ∈ (0, 1).**

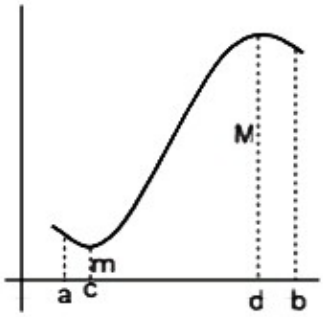
Sol. Let g(x) = f(x) + 2x² – 1 is a continuous function in (0, 1).

g(0) = f(0) – 1 = -1 and g(1) = f(1) + 1 = 1 then there exists some c ∈ (0, 1) such that g(c) = 0.

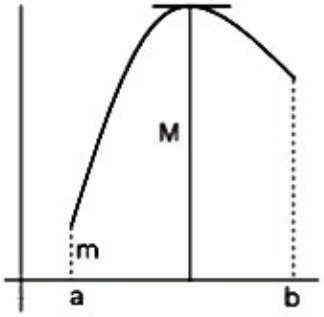
f(c) = 1 – 2c².

T-3 : Extreme value Theorem :

If f is continuous on [a, b] then f takes on, a least value 'm' and a greatest value 'M' on this interval.



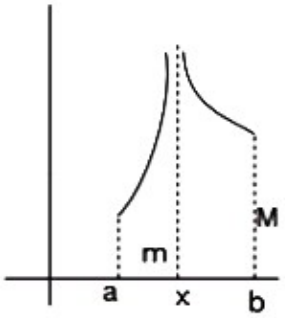
Minimum value 'm' occurs at x=c and maximum value M occurs at x=d where c, d∈(a, b)



Minimum value 'm' occurs at the end point x=a and maximum value M occurs inside the interval

Note :

To see that continuity is necessary for the extreme value theorem to be true refer the graph shown.
There is a discontinuity at x = c. The function has a minimum value at the left end point x = a and f has no maximum value.



PROPERTIES OF FUNCTION CONTINUOUS IN [a, b]:

(i) If a function f is continuous on a closed interval [a, b] then it is bounded.

(ii) A continuous function whose domain is some closed interval must have its range also in closed interval.

(iii) If f is continuous and onto on [a, b] and is one-one then f^{-1} (from the range of f) is also continuous.

(iv) If f(a) and f(b) posses opposite signs then ∃ at least one solution of the equation f(x) = 0 in the open interval (a, b) provided f is continuous in [a, b].

Q.1 **Let f be a continuous function defined onto on [0, 1] with range [0, 1]. Show that there is some c in (0, 1) such that f(c) = 1 – c.**

Sol. Consider $g(x) = f(x) - 1 + x$ $\quad\because$ g(x) is continuous in [0, 1]

$g(0) = f(0) - 1 \le 0$ [as $f(0) \le 1$]

$g(1) = f(1) \ge 0$ [as $f(1) \ge 0$]

Hence, g(0) and g(1) have values of opposite signs.

Hence, there exists at least one $c \in (0, 1)$ such that g(c) = 0.

$\therefore$ $g(c) = f(c) - 1 + c = 0$; $f(c) = 1 - c$.

Q.2 **Let f be continuous on the interval [0, 1] to R such that f(0) = f(1). Prove that there exists a point c in $\left[0, \frac{1}{2}\right]$ such that $f(c) = f\left(c + \frac{1}{2}\right)$.**

Sol. Consider a continuous function $g(x) = f\left(x + \frac{1}{2}\right) - f(x)$

$\left(g \text{ is continuous } x \in \left[0, \frac{1}{2}\right]\right)$

$\Rightarrow \quad g(0) = f\left(\frac{1}{2}\right) - f(0) = f\left(\frac{1}{2}\right) - f(1)$ [as f(0) = f(1)]

and $g\left(\frac{1}{2}\right) = f(1) - f\left(\frac{1}{2}\right) = -\left[f\left(\frac{1}{2}\right) - f(1)\right]$

Since g is continuous and g(0) and $g\left(\frac{1}{2}\right)$ have opposite signs, hence the equation g(x) = 0 must have atleast one root in $\left[0, \frac{1}{2}\right]$.

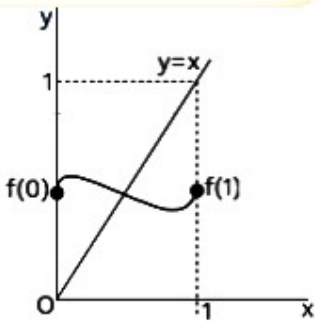
Hence, for some $c \in \left[0, \frac{1}{2}\right]$, $g(c) = 0 \Rightarrow f\left(c + \frac{1}{2}\right) = f(c)$.

Q.3 **Let $f : [0, 1] \to [0, 1]$ be a continuous function. Then prove f(x) = x for at least one x ∈ [0, 1].**

Sol. Clearly, $0 \le f(0) \le 1$ and $0 \le f(1) \le 1$.

As f(x) is continuous, f(x) attains all the values between f(0) and f(1) and the graph will have no breaks.

So, the graph will cut the line y = x atleast at one point x where $0 \le x \le 1$. So, f(x) = x at that point.



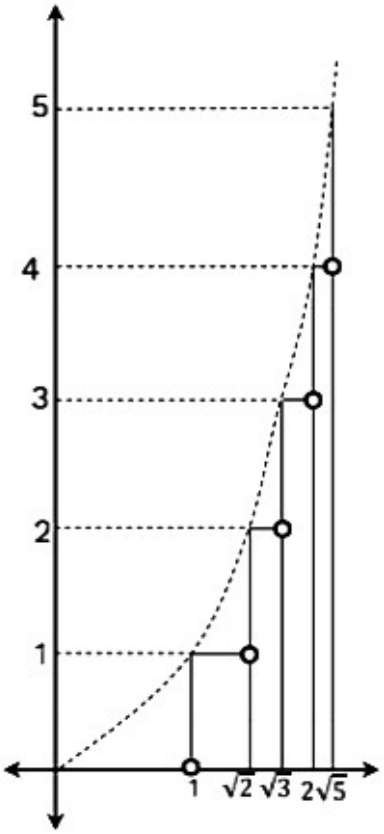
CONTINUITY OF SPECIAL TYPES OF FUNCTIONS:

Continuity Of Functions In Which Greatest Integer Function Is Involved :

$f(x) = [x]$ is discontinuous when x is an integer.

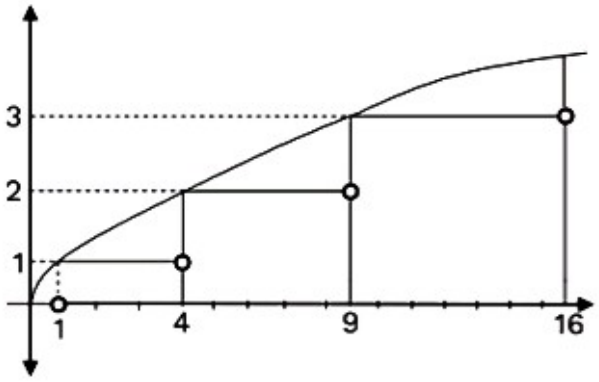
Similarly, $f(x) = [g(x)]$ is discontinuous when $g(x)$ is an integer, but this is true only when $g(x)$ is monotonic ($g(x)$ is strictly increasing or strictly decreasing).

For example, $f(x) = \left[\sqrt{x}\right]$ is discontinuous when $\sqrt{x}$ is an integer as $\sqrt{x}$ strictly increasing (monotonic function).

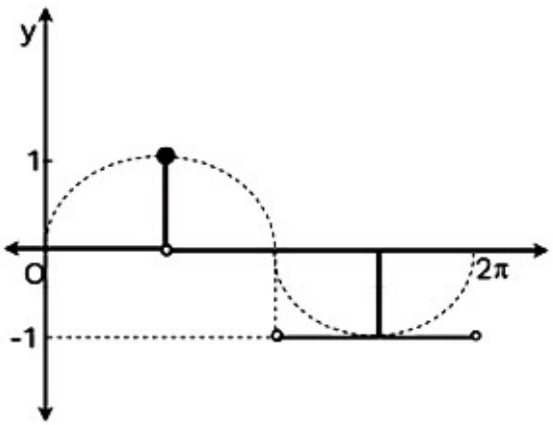


$f(x) = [x^2]$, $x \geq 0$ is discontinuous when x^2 is an integer, as x^2 is strictly increasing for $x \geq 0$.

Now consider, $f(x)=[\sin x]$, $x \in [0, 2\pi]$, $g(x)=\sin x$ is not monotonic in $[0, 2\pi]$.



For this type of function, points of discontinuity can determined by graphical methods. We can note that at $x = \frac{3\pi}{2}$, sinx takes integral value -1, but at $x = \frac{3\pi}{2}$, $f(x) = [\sin x]$ is continuous.



Q.1 **Discuss the continuity of following functions ([.] represents the greatest integer function.**

(i) $f(x) = [\log_e x]$

Sol. $\log_e x$ function is a monotonically increasing function.
Hence $f(x) = [\log_e x]$ is discontinuous, where $\log_e x = k$ or $x = e^k$, $k \in Z$

Thus f(x) is discontinuous at $x = ...e^{-2}, e^{-1}, e^0, e^1, e^2, ...$

Q. **(ii) $f(x) = [\sin^{-1}x]$**

Sol. $\sin^{-1}x$ is a monotonically increasing function.

Hence, $f(x) = [\sin^{-1}x]$ is discontinuous where $\sin^{-1}x$ is an integer.
$\Rightarrow \sin^{-1}x = -1, 0, 1$ or $x = -\sin 1, 0, \sin 1$

Q. **(iii)** $f(x) = \left[\frac{2}{1+x^2}\right], x \geq 0$

Sol. $\frac{2}{1+x^2}$, $x \geq 0$, is a monotonically decreasing function.

Hence, $f(x) = \left[\frac{2}{1+x^2}\right], x \geq 0$ is discontinuous, when $\frac{2}{1+x^2}$ is an integer.

$\Rightarrow \frac{2}{1+x^2} = 1, 2$

$\Rightarrow x = 1, 0$

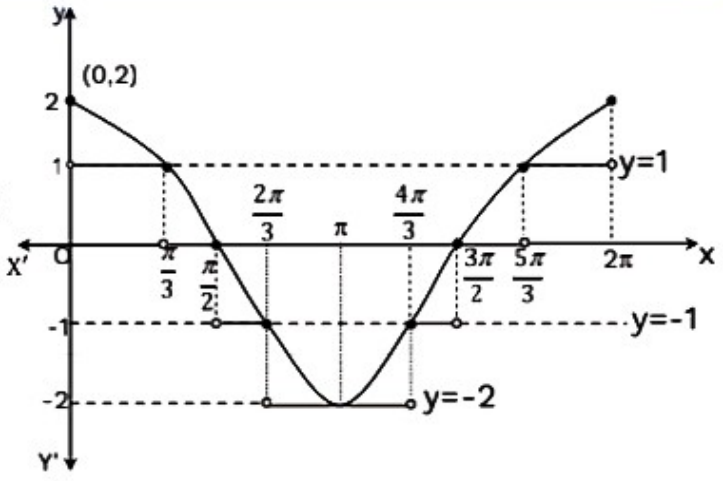
Q.2 **Draw the graph and find the points of discontinuity for $f(x) = [2\cos x]$, $x \in [0, 2\pi]$, ([.] represents the greatest integer function).**

Sol. $f(x) = [2\cos x]$

Clearly from the graph given in figure f(x) is discontinuous at points where, $2\cos x = \pm 1, 0, \pm 2$

from graph

$x = 0, \frac{\pi}{3}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{3\pi}{2}, \frac{5\pi}{3}, 2\pi$



Q.3 **Draw the graph and discuss the continuity of f(x) = [sinx + cosx], x ∈ [0, 2π], where [.] represents the greatest integer function.**

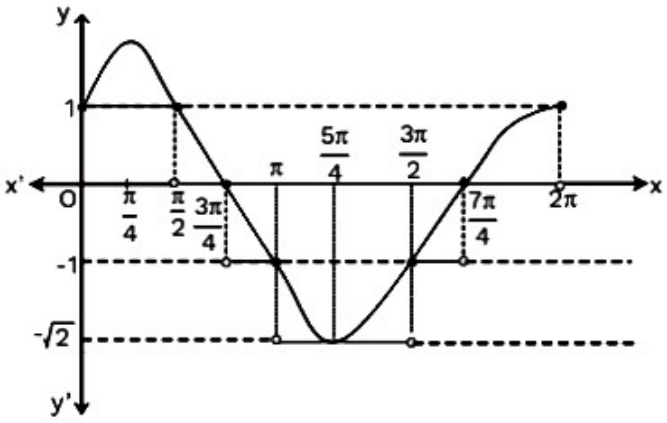
Sol. f(x) = [sinx + cosx] = [g(x)] where g(x) = sinx + cosx

$$g(0)=1, g\left(\frac{\pi}{4}\right)=\sqrt{2}, g\left(\frac{\pi}{2}\right)=1$$

$$g\left(\frac{3\pi}{4}\right)=0, g(\pi)=-1, g\left(\frac{5\pi}{4}\right)=-\sqrt{2}$$

$$g\left(\frac{3\pi}{2}\right)=-1,\ g\left(\frac{7\pi}{4}\right)=0, g(2\pi)=1$$

Clearly from the graph given in figure f(x) is discontinuous at $x=\frac{\pi}{2}, \frac{3\pi}{4}, \pi, \frac{3\pi}{2}, \frac{7\pi}{4}, 2\pi$.



Q.4 **If the function $f(x)=\left[\frac{(x-2)^3}{a}\right]\sin(x-2)+a\cos(x-2)$, ([.] denotes the greatest integer function) which is continuous in [4, 6], then find the value of a.**

Sol. sin(x - 2) and cos(x - 2) are continuous for all x.

Since [x] is not continuous at integer point.

So, f(x) is continuous in [4, 6] if $\left[\frac{(x-2)^3}{a}\right]=0\ \forall\ x\in[4,6]$.

Now $(x-2)^3\in[8,64]$ for $x\in[4,6]$.

$\Rightarrow\ a>64$ for $\left[\frac{(x-2)^3}{a}\right]=0$

Continuity Of Functions Is Which Signum Function Is Involved :

We know that f(x) = sgn (x) is discontinuous at x = 0.

In general, f(x) = sgn(g(x)) is discontinuous at x = a if g(a) = 0.

Q.5 **Discuss the continuity of**

(i) $f(x) = \text{sgn}(x^3 - x)$

Sol. $f(x) = \text{sgn}(x^3 - x)$

Here $x^3 - x = 0 \Rightarrow x = 0, -1, 1$

Here f(x) is discontinuous at x=0, -1, 1

Q. **(ii) f(x) = sgn(2cosx – 1)**

Sol. f(x) = sgn(2cosx – 1)

Here, 2 cos x – 1 = 0

$\Rightarrow \cos x = \frac{1}{2}$

$\Rightarrow x = 2n\pi \pm \left(\frac{\pi}{3}\right)$

$n \in Z$, where f(x) is discontinuous.

Q. **(iii)** $f(x) = \text{sgn}(x^2 - 2x + 3)$

Sol. $f(x) = \text{sgn}(x^2 - 2x + 3)$

Here, $x^2 - 2x + 3 > 0$ for all x.

Thus, f(x) = 1 for all x, hence continuous for all x.

Q.6 **If f(x) = sgn(2sinx + a) is continuous for all x, then find the possible values of a.**

Sol. f(x) = sgn(2sinx + a) is continuous for all x.

Then 2 sin x + a ≠ 0 for any real x.

$\Rightarrow \sin x \neq -\frac{a}{2}$

$\Rightarrow \left|\frac{a}{2}\right| > 1 \Rightarrow a < -2 \text{ or } a > 2$

Continuity of Functions Involving Limit $\lim_{n\to\infty} a^n$:

We know that $\lim_{n\to\infty} a^n = \begin{cases} 0, & 0 \le a < 1 \\ 1, & a = 1 \\ \infty, & a > 1 \end{cases}$

Q.1 **Discuss the continuity of $f(x) = \lim_{n\to\infty} \frac{x^{2n}-1}{x^{2n}+1}$.**

Sol. $f(x) = \lim_{n\to\infty} \frac{x^{2n}-1}{x^{2n}+1} = \lim_{n\to\infty} \frac{1-\frac{1}{(x^2)^n}}{1+\frac{1}{(x^2)^n}} = \begin{cases} -1, & 0 \le x^2 < 1 \\ 0, & x^2 = 1 \\ 1, & x^2 > 1 \end{cases} \begin{cases} 1, & x < -1 \\ 0, & x = -1 \\ -1, & -1 < x < 1 \\ 0, & x = 1 \\ 1, & x > 1 \end{cases}$

Thus, f(x) is discontinuous at x = ± 1.

Q.2 **Discuss the continuity of $f(x) = \lim_{n\to\infty} \cos^{2n} x$.**

Sol. $f(x) = \lim_{n\to\infty} (\cos^2 x)^n$

$= \begin{cases} 0, & 0 \le \cos^2 x < 1 \\ 1, & \cos^2 x = 1 \end{cases} = \begin{cases} 0, & x \neq n\pi, n \in I \\ 1, & x = n\pi, n \in I \end{cases}$

Hence, f(x) is discontinuous when x = nπ, n ∈ I.

Functions Continuous Only At Some Points And Defined Everywhere (Discrete Point Continuity):

Q.1 Discuss the continuity for $f(x) = \begin{cases} x, & \text{if } x \in Q \\ 0, & \text{if } x \notin Q \end{cases}$.

Sol. If $x \to 0^+$ then x may be rational or irrational

$\Rightarrow f(0^+) = 0\,(x \in Q)$ or $f(0^+) = 0\,(x \notin Q)$

If $x \to 0^-$ then x may be rational or irrational

$\Rightarrow f(0^-) = 0\,(x \in Q)$ or $f(0^-) = 0\,(x \notin Q)$

Hence f(x) is continuous at x = 0.

Q.2 **Discuss the continuity for** $f(x) = \begin{cases} x, & \text{if } x \in Q \\ -x, & \text{if } x \notin Q \end{cases}$.

Sol. f(x) is continuous at some x = a, where x = – x or x = 0.

If $x \to 0^+$ then x may be rational or irrational

$\Rightarrow f(0^+) = 0\,(x \in Q)$ or $f(0^+) = -0 = 0\,(x \notin Q)$

If $x \to 0^-$ then x may be rational or irrational

$\Rightarrow f(0^-) = 0\,(x \in Q)$ or $f(0^-) = -0 = 0\,(x \notin Q)$

Hence f(x) is continuous at x = 0.

Q.3 **Discuss the continuity for** $f(x) = \begin{cases} x, & \text{if } x \in Q \\ 1-x, & \text{if } x \notin Q \end{cases}$.

Sol. f(x) is continuous at some x = a, where x = 1 - x or $x = \frac{1}{2}$.

If $x \to \frac{1}{2^+}$ then x may be rational or irrational

$\Rightarrow f\left(\frac{1}{2^+}\right) = \frac{1}{2}$ or $1 - \frac{1}{2} = \frac{1}{2}$

If $x \to \frac{1}{2^-}$ then x may be rational or irrational

$\Rightarrow f\left(\frac{1}{2^-}\right) = \frac{1}{2}$ or $1 - \frac{1}{2} = \frac{1}{2}$

Hence f(x) is continuous at $x = \frac{1}{2}$

For some other point, say, x = 1

$\Rightarrow$ f(1) = 1

If $x \to 1$ then x may be rational or irrational.

$\Rightarrow$ f(1) = 1 or 1 – 1 = 0

Hence, f(1) oscillates between 1 and 0, which causes discontinuity at x = 1.

Similarly, f(x) oscillates between two numbers for all $x \in R - \left(\frac{1}{2}\right)$.

Q.4 **Discuss the continuity for** $f(x) = \begin{cases} x^2, & \text{if } x \in Q \\ 1, & \text{if } x \notin Q \end{cases}$.

Sol. f(x) is continuous at some x = a, where $x^2 = 1$ or $x = \pm 1$.

For x = 1,

If $x \to 1^+$ then x may be rational or irrational

$\Rightarrow f(1^+) = (1)^2 = 1 (x \in Q)$ or $f(1^+) = 1 \ (x \notin Q)$

If $x \to 1^-$ then x may be rational or irrational

$\Rightarrow f(1^-) = (1)^2 \ (x \in Q)$ or $f(1^-) = 1 \ (x \notin Q)$

Hence, f(x) is continuous at x = 1.

For x = –1,

If $x \to (-1^+)$ then x may be rational or irrational

$\Rightarrow f(-1^+) = (-1)^2 = 1 (x \in Q)$ or $f(-1^+) = 1 \ (x \notin Q)$

If $x \to (-1^-)$ then x may be rational or irrational

$\Rightarrow f(-1^-) = (-1)^2 \ (x \in Q)$ or $f(-1^-) = 1 \ (x \notin Q)$

Hence, f(x) is continuous at x = –1.

Continuity of Composite Functions :

If f is continuous at x = a and g is continuous at x = f(a) then the composite function g[f(x)] is continuous at x = a.

e.g. $f(x)=\frac{x\sin x}{x^2+2}$ and g(x) = |x| are continuous at x = 0, hence the composite $gof(x)=\left|\frac{x\sin x}{x^2+2}\right|$ will also be continuous at x = 0.

Q.1 **If $f(x)=\frac{x+1}{x-1}$ and $g(x)=\frac{1}{x-2}$, then discuss the continuity of f(x), g(x) and fog(x).**

Sol. **(a)** $f(x)=\frac{x+1}{x-1}$

$\therefore$ f(x) is not defined at x = 1. $\therefore$ f is discontinuous at x = 1.

(b) $g(x)=\frac{1}{x-2}$

g(x) is not defined at x = 2. $\therefore$ g is discontinuous at x = 2

(c) Now, fog will be discontinuous at x = 2 [point of discontinuity of g(x)] and when g(x) = 1 (when value of g(x) is point of discontinuity of f(x))

If $g(x)=1 \Rightarrow \frac{1}{x-2}=1 \Rightarrow x=3$.

$\therefore$ fog (x) is discontinuous at x=2 and x=3.

Also, $fog(x)=\dfrac{\frac{1}{x-2}+1}{\frac{1}{x-2}-1}$

Here fog (2) is not defined.

$$\lim_{x\to 2} fog(x)=\lim_{x\to 2}\frac{\frac{1}{x-2}+1}{\frac{1}{x-2}-1}=\lim_{x\to 2}\frac{1+x-2}{1-x+2}=1$$

$\therefore$ fog(x) is discontinuous at x = 2 and it has a removable discontinuity at x=2.
For checking continuity at x = 3.

$$\lim_{x\to 3^+} fog(x) = \lim_{x\to 3^+} \frac{\frac{1}{x-2}+1}{\frac{1}{x-2}-1} = -\infty$$

$$\lim_{x\to 3^-} fog(x) = \lim_{x\to 3^-} \frac{\frac{1}{x-2}+1}{\frac{1}{x-2}-1} = +\infty$$

$\therefore$ fog(x) is discontinuous at x = 3 and has non removable discontinuity at x = 3.